

Books and References:

1. Montgomery, D. C., Peck, E. A., & Vining, G. G. (2012). Introduction to Linear Regression Analysis. Wiley.
2. D. D Joshi (1987). Linear Estimation and Design of Experiments. Wiley
3. Darlington, R. B. (1990). Linear Regression Analysis: Assumptions and Applications. Sage Publications.
4. Seber, G. A. F., & Lee, A. J. (2003). Linear Regression Analysis. Wiley
5. Weisberg, S. (2014). Applied Linear Regression. Wiley.
6. Yan, X., & Chen, M. (2007). Linear Regression Analysis: Theory and Computing. World Scientific

Programme	B. Sc. Statistics				
Course Code	STA6CJ302(P)				
Course Title	Design and Analysis of Experiments				
Type of Course	Major				
Semester	VI				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites					
Course Summary Objective	Objective make students aware of designing, planning conducting analysing interpreting controlled tests, analysing. Differentiating the variation from various sources				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Identify various causes of variation	U	C	Instructor-created exams / Quiz
CO2	Describe models to express the resulting observation based effects and errors	Ap	P	Practical Assignment / Observation of Practical Skills
CO3	Discuss the fundamental principles of	Ap	P	Seminar Presentation /

	experiments			Group Tutorial Work
CO4	Summarize the total variation into sum of fixed and random causes	U	C	Instructor-created exams / Home Assignments
CO5		Ap	P	One Minute Reflection Writing assignments
CO6		Ap	P	Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	Theory of Linear Estimation		10	15
	1	Estimability of linear parametric functions.		
	2	Method of least squares		
	3	Best Linear Unbiased Estimator (BLUE)		
	4	Gauss -Markov theorem		
	5	Linear hypothesis, Estimation of error variance.		
	Sections from References:			
II	Analysis of variance		10	15
	6	Definitions of Fixed effect model and random effect models		
	7	Definition of analysis of Variance,		
	8	Assumptions and Limitations of ANOVA		
	9	One way ANOVA with a single observation per cell		
	10	Two-way ANOVA with a single observation per cell		
	Sections from References:			
III	Analysis of covariance		17	20
	11	Model of Analysis of covariance		
	12	Analysis of covariance with a single observation per cell		
	13	Experimental Designs		
	14	Terminology Unit, Material Treatment and Experimental error.		
	15	Principles of design of Experiment		
	16	Randomization		
	17	Replication		
	18	Local Control		
	Sections from References:			
IV	Basic Designs		8	20
	19	Completely randomized design (CRD)		